Numerical Analysis – Homework #2 (Solutions)

Utah State University Fall 2025

1.

Consider the system of linear equations:

$$\begin{aligned} x_1 + 2x_2 &= -1, \\ -2x_1 - 3x_2 + 2x_3 &= 3, \\ -2x_2 - 3x_3 &= -3. \end{aligned}$$

(a) Write the system as $Ax = b$.

(b) Solve the system using Gaussian Elimination.

(c) Find the LU decomposition of A and solve using it.

(d) Compute A^{-1}.

(e) What is the determinant of A?

(f) What is the condition number of A?

Solution.

$$A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -3 & 2 \\ 0 & -2 & -3 \end{bmatrix}, \quad b = \begin{bmatrix} -1 \\ 3 \\ -3 \end{bmatrix}.$$

Gaussian elimination gives upper-triangular $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$, leading to $(x_1, x_2, x_3) = (-7, 3, -1)$.

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$A^{-1} = \begin{bmatrix} 13 & 6 & 4 \\ -6 & -3 & -2 \\ 4 & 2 & 1 \end{bmatrix}, \quad \det(A) = 1, \quad \kappa_1(A) = 7 \times 23 = 161.$$

2.

Evaluate the norms $\| * \|_p$ where $p = 1, \infty$ for the following.

(a)

$$A = \begin{bmatrix} -9 & 5 & -9 \\ -2 & 7 & 5 \\ 5 & 1 & 8 \end{bmatrix}$$

(b)

$$B = \begin{bmatrix} 8 & -2 & 1 \\ -2 & 2 & -1 \\ -2 & 4 & -3 \end{bmatrix}$$

(c) Compute B^2, $A+B$, and AB. Does the triangle inequality hold?

Solution.

$$\|A\|_1 = 22, \quad \|A\|_\infty = 23; \qquad \|B\|_1 = 12, \quad \|B\|_\infty = 11.$$

Triangle inequality holds since $\|A+B\|_1 = 17 \le 34$, $\|A+B\|_\infty = 17 \le 34$.

3.

Let A and B be two $n \times n$ matrices. Show that the number of nonzero scalar multiplications required to compute $C = AB$ is

$$\sum_{i=1}^{n} \operatorname{nnz}\big(A(:,i)\big) * \operatorname{nnz}\big(B(i,:)\big).$$

Solution. Each column i of A multiplies each row i of B; only nonzero entries contribute. Hence total multiplications $= \sum_i \operatorname{nnz}(A(:,i))\operatorname{nnz}(B(i,:))$.

4.

Consider

$$A(\theta) = \begin{pmatrix} 1 & \cos\theta \\ \cos\theta & 1 \end{pmatrix}.$$

(a) Write $\kappa_1(A(\theta)) = \|A\|_1 \|A^{-1}\|_1$ as a function of θ.

(b) For $\theta = 10^{-5}$, estimate the relative error bound and expected digits of accuracy (16-digit arithmetic).

Solution. Let $c = \cos\theta$.

$$A^{-1} = \frac{1}{1-c^2}\begin{bmatrix} 1 & -c \\ -c & 1 \end{bmatrix}, \quad \|A\|_1 = 1 + |c|, \quad \|A^{-1}\|_1 = \tfrac{1+|c|}{1-c^2}.$$

$$\Rightarrow \ \kappa_1(A) = \frac{(1+c)^2}{1-c^2} = \frac{1+c}{1-c} \approx \frac{4}{\theta^2}.$$

For $\theta = 10^{-5}$, $\kappa \approx 4 \times 10^{10}$; error $\lesssim \kappa\varepsilon \approx 4 \times 10^{-6}$, so about 5–6 accurate digits.

5.

You want to solve $Ax = b$, where

$$A = \begin{pmatrix} 1.01 & 1 \\ 1 & 1.01 \end{pmatrix}.$$

b has only $\hat{b}$ accurate to about three digits. If $A\hat{x} = \hat{b}$, find an upper bound on $\|\hat{x}-x\|/\|x\|$ using the perturbation theorem.

Solution.

$$\frac{\|\hat{x}-x\|}{\|x\|} \le \kappa(A)\frac{\|\delta b\|}{\|b\|}.$$

A has $\kappa_2(A) \approx 201$. If $\|\delta b\|/\|b\| \approx 10^{-3}$, then $\frac{\|\hat{x}-x\|}{\|x\|} \lesssim 0.2$ ($\approx 20\%$ possible error).

6.

Suppose

$$A = \begin{pmatrix} 1 & 0 & \cdots & 0 & 1 \\ -1 & 1 & \ddots & \vdots & \vdots \\ \vdots & -1 & \ddots & 0 & 1 \\ \vdots & \vdots & \ddots & 1 & 1 \\ -1 & -1 & \cdots & -1 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ \vdots \\ 1 \\ 2 \\ 1 \end{pmatrix}.$$

Gaussian elimination gives

$$U = \begin{pmatrix} 1 & 0 & \cdots & 0 & 1 \\ 0 & 1 & \ddots & \vdots & 2 \\ \vdots & 0 & \ddots & 0 & \vdots \\ \vdots & \vdots & \ddots & 1 & 2^{n-2} \\ 0 & 0 & \cdots & 0 & 2^{n-1} \end{pmatrix}, \; c = \begin{pmatrix} 1 \\ 2 \\ \vdots \\ 2^{n-2}+1 \\ 2^{n-1}+1 \end{pmatrix}.$$

Solution. If $2^{n-2} > 1/\epsilon$, rounding causes $2^k + 1 \to 2^k$. Back substitution gives $\hat{x}_n = 1$, all others 0:

$$\boxed{\hat{x} = (0, \ldots, 0, 1)^T}.$$

7.

Surveying altitudes problem:

$$\begin{aligned} x_1 &= 2.95, & x_2 &= 2.95, & x_3 &= -1.45, & x_4 &= -1.45, \\ x_1 - x_2 &= 1.23, & x_1 - x_3 &= 4.45, & x_1 - x_4 &= 1.61, \\ & & x_2 - x_3 &= 3.21, & x_2 - x_4 &= 0.45, \\ & & & & x_3 - x_4 &= -2.75. \end{aligned}$$

Find the least-squares solution via normal equations and Cholesky.

Solution.

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}, \quad b = \begin{bmatrix} 2.95 \\ 2.95 \\ -1.45 \\ -1.45 \\ 1.23 \\ 4.45 \\ 1.61 \\ 3.21 \\ 0.45 \\ -2.75 \end{bmatrix}.$$

$$A^T A = \begin{bmatrix} 4 & -1 & -1 & -1 \\ -1 & 4 & -1 & -1 \\ -1 & -1 & 4 & -1 \\ -1 & -1 & -1 & 4 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 10.24 \\ 5.38 \\ -11.86 \\ -0.76 \end{bmatrix}.$$

Solve $(A^TA)x = A^Tb$ via Cholesky:

$$x \approx (2.65,\ 1.68,\ -1.77,\ 0.45)^T, \quad \kappa_2(A^TA) \approx 5.$$

Since κ is small, normal equations are numerically safe.